Technical Service Bulletin

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Coolant Level Sensor Leakage Issues	

Coolant Level Sensor Leakage Issues

Core Issue

This Technical Service Bulletin describes a condition in which the coolant level sensor can malfunction, allowing coolant to seep into the wiring harness and into other sensors and electronic control system devices. Both two wire and three wire coolant level sensors have been observed to leak internally. 2007 and 2010 EPA-certified automotive engines built with an aftertreatment system can experience white smoke in extreme cases where coolant seeps through the wiring harness to the aftertreatment diesel particulate filter differential pressure sensor.

Confirmation

Automotive, Industrial, and Marine applications are affected.

- ISB CM2250
- ISB CM2150
- ISB CM850
- QSB CM850
- ISC CM2250
- ISC CM2150
- ISC CM850
- QSC CM850
- ISL CM2250
- ISL CM2150
- ISL CM850
- QSL CM850
- ISM CM876
- ISM CM875
- QSM CM570
- ISX CM2250

- ISX CM871
- ISX CM870
- QSX CM570

Symptoms can vary by engine family, OEM, and application. Symptoms include, but are **not** limited to:

- Slow coolant consumption.
- Coolant level sensor fault codes. The engine ECM does **not** always log a fault code for the coolant level sensor when it begins to leak coolant internally.
- Electronic system component corrosion.
- Electronic system component electrical connections wet with coolant.
- White smoke. This **only** occurs on engines built with an aftertreatment system and an aftertreatment diesel particulate filter differential pressure sensor.

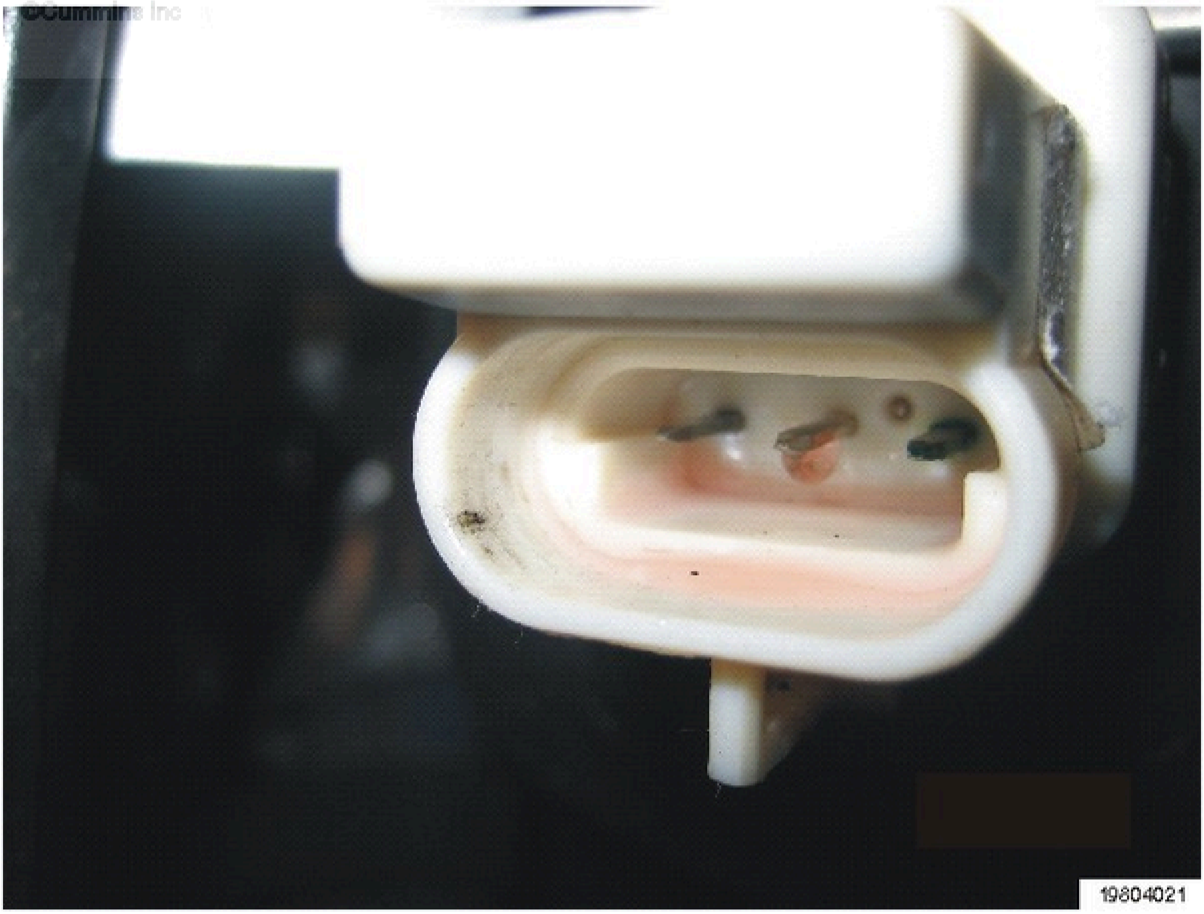
Disconnect the wiring harness from the coolant level sensor and inspect the male and female connections for evidence of coolant intrusion. The electrical connection is a tight seal and an external leak will **not** be evident. The same should also be done to the aftertreatment sensor connectors.

The coolant level sensor is leaking internally. Current coolant level sensors are manufactured with materials that can **not** withstand exposure to extended life coolant.

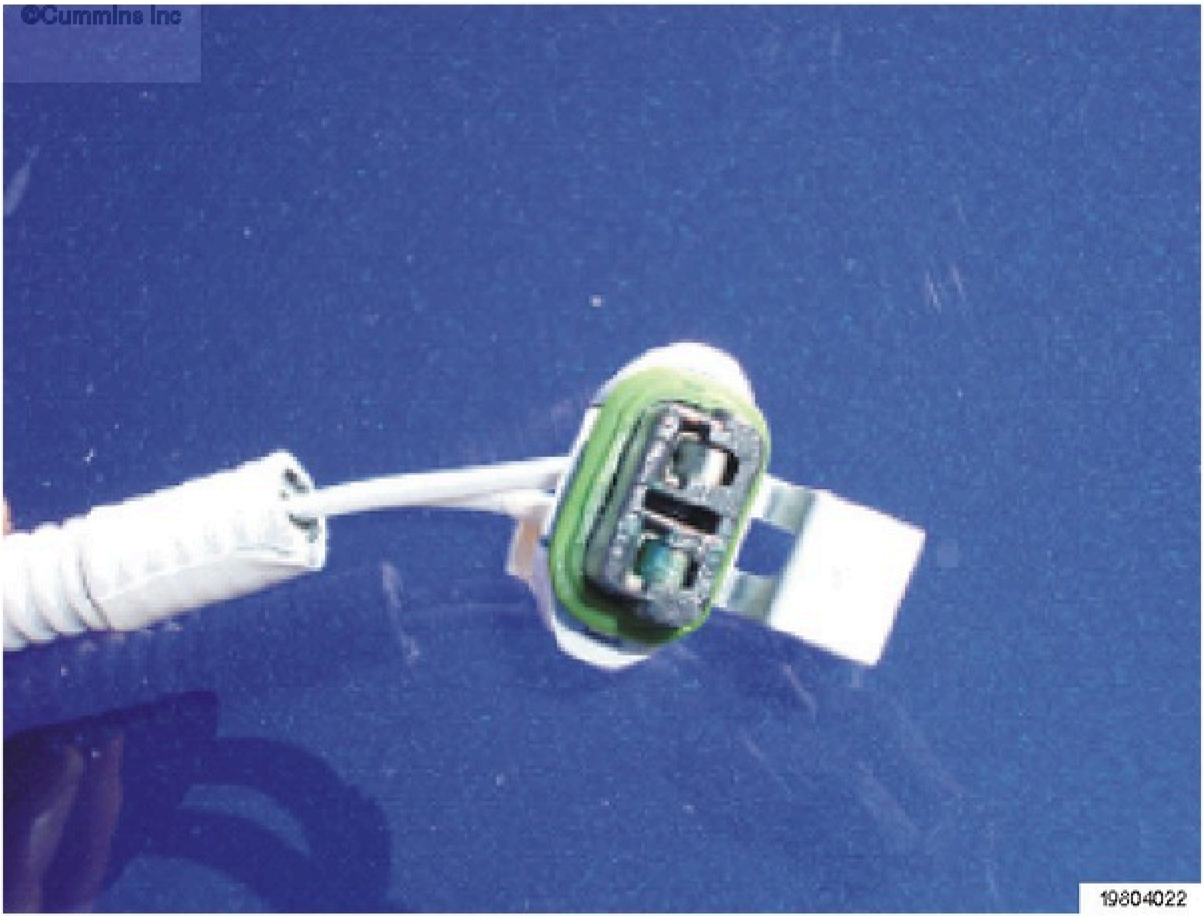
Extended operation with a leaking coolant level sensor will result in total contamination of the engine wiring harness. It will be necessary to disconnect the harness connectors, including the OEM 60-pin ECM connector, to inspect for coolant intrusion. Replace any damaged components.

Resolution

Coolant level sensors are commonly OEM-supplied. Depending upon vehicle application, it is possible that the sensor was supplied by Cummins Inc. The simplest method to determine who supplied the sensor, is to view the part number stamped on the sensor and determine if that part number is a Cummins part number or an OEM part number. If the sensor is OEM-supplied, work with the vehicle OEM to replace any damaged harnesses, sensors, ECMs, or other control modules that are part of the electronic control system. This is **not** a failure warrantable by Cummins® Inc. Repair locations should work with the vehicle OEM for warranty reimbursement.



Coolant intrusion in the coolant sensor.



A corroded electrical connection that requires replacement.

Warranty Statement

The information in this document has no effect on present warranty coverage or repair practices, nor does it authorize TRP or Campaign actions.

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